

1. Administrative & Study Identification

Full Study Title: A randomized, open-label, multi-arm, two-part, phase II study to assess the efficacy and safety of multiple LXH254 combinations in patients with previously treated unresectable or metastatic BRAFV600 or NRAS mutant melanoma **Short Title/Acronym:** Study of Efficacy and Safety of LXH254 Combinations in Patients With Previously Treated Unresectable or Metastatic Melanoma **Protocol Number:** CLXH254C12201 (EudraCT: 2020-000873-26) **NCT Number:** NCT04417621 **Posting ID:** 1035438760563737702 **Sponsor & Company Name:** Novartis (Novartis Pharmaceuticals / Novartis Pharma AG) **Investigational Product:** LXH254 (naporafenib) in combination with LTT462, trametinib, or ribociclib **Phase of Development:** Phase II (PHASE2) **Trial Status:** ACTIVE_NOT_RECRUITING (Active but not currently recruiting) **Medical Monitor:** Novartis Pharmaceuticals Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy, specifically the Overall Response Rate (ORR), of LXH254 (naporafenib) combinations in patients with previously treated unresectable or metastatic BRAFV600 or NRAS mutant melanoma. **Secondary Objectives:** To assess the safety, tolerability, Duration of Response (DOR), Progression-Free Survival (PFS), and Disease Control Rate (DCR) of the combination therapies.

2.2 Background & Rationale To address the critical unmet medical need in patients with advanced melanoma harboring BRAFV600 or NRAS mutations who have progressed on prior therapies. While targeted therapies and immune checkpoint inhibitors (CPI) have transformed melanoma treatment, intrinsic and acquired resistance frequently occurs, often driven by the reactivation of the MAPK pathway. LXH254 is a potent, selective pan-RAF inhibitor. Combining LXH254 with other targeted agents (trametinib, ribociclib, or LTT462) aims to achieve deeper and more durable MAPK pathway inhibition, overcoming resistance and preventing compensatory signaling pathways.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator / Multi-arm **Blinding:** Open-label **Randomization:** Randomized (Multi-arm allocation)

3.2 Planned Enrollment & Duration Targeted Enrollment (\$N\$): 134 participants **Number of Sites:** Global, Multicenter **Estimated Study Start:** 29-Oct-2020 **Estimated Primary Completion:** 27-Mar-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male or female ≥ 12 years of age (adolescents 12-17 years must have a body weight > 40 kg). 2. Histologically confirmed unresectable or metastatic cutaneous melanoma with NRAS or BRAFV600 mutation. 3. *For NRAS mutation:* Must have received prior systemic therapy with checkpoint inhibitors (CPI), either an anti-PD-1/PD-L1 as a single agent or in combination with anti-CTLA-4, investigational agents, chemotherapy or locally directed anti-neoplastic agents (max 2 prior lines of CPI-containing immunotherapy) with documented progressive disease. Confirmation of progressive disease is not required for patients who remained on treatment for > 6 months. *For BRAFV600 mutation:* Must have received prior CPI therapy AND targeted therapy with a RAF inhibitor single agent or in combination with a MEK inhibitor (+/- CPI allowed) as the last prior therapy. Maximum of one line of targeted therapy is allowed. Must have documented progressive disease as per RECIST v1.1 while on/after treatment.

4.2 Exclusion Criteria 1. Treatment with specific anti-cancer therapies prior to the first dose within stated timeframes (≤ 4 weeks for radiation therapy, immunotherapy including CPIs, chemotherapy, locally directed anti-neoplastic agents, or other investigational agents; ≤ 2 weeks for limited field radiation for palliation or small molecule therapeutics; ≤ 6 weeks for cytotoxic agents with major delayed toxicities). 2. Primary central nervous system (CNS) tumors or symptomatic CNS metastases that are neurologically unstable. 3. History or current evidence of retinal vein occlusion (RVO) or current risk factors for RVO (e.g., uncontrolled glaucoma, ocular hypertension, history of hyperviscosity or hypercoagulability syndromes). 4. Participants participating in additional parallel investigational drug or medical device studies. 5. Any medical condition that would prevent the patient's participation due to safety concerns or compliance.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: LXH254 administered in combination with either LTT462 (ERK inhibitor), trametinib, or ribociclib (CDK4/6 inhibitor). *Drug Profile for Trametinib:* Trade name: Mekinist | ATC Code: L01EE01 | [EMA Product Information](#) Dosages are determined by protocol arms and continuous 28-day cycles. **Route of Administration:** Oral (all investigational study drugs are formulated as pills). **Duration of Treatment:** Continued

until disease progression, unacceptable toxicity, or withdrawal of consent.

5.2 Concomitant & Prohibited Therapy Permitted: Routine supportive care. Additional agents administered with CPI during the previous lines of therapy are permitted. **Prohibited:** Participants participating in additional parallel investigational drug or medical device studies are excluded.

6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :---
| :--- | :--- | | **Screening** | Day -28 to 0 | Informed Consent, Medical History, Histological confirmation, Baseline Tumor Assessment | | **Baseline** | Day 0 | Randomization, First Dose | | **Treatment (Cycle 1 & 2)** | Weeks 1-8 | 3 clinic visits per cycle: Safety Labs, Pharmacokinetics, Efficacy Assessment | | **Treatment (Cycle 3+)** | Weeks 9+ | 2 visits during Cycle 3; 1 visit per cycle thereafter: Continual Safety & Efficacy evaluations | | **End of Study** | Follow-up | 30-day safety follow-up / 120-day pregnancy follow-up; Telephone survival follow-up up to 4 times a year |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Overall Response Rate (ORR)
Measurement Method: Confirmed ORR assessed per RECIST v1.1 using local investigator assessment.

7.2 Secondary & Exploratory Endpoints 1. Duration of Response (DOR) 2. Progression-Free Survival (PFS) 3. Disease Control Rate (DCR) 4. Safety and Tolerability (Incidence and severity of adverse events)

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring throughout the study via clinical visits and laboratory assessments. High surveillance for skin toxicities (e.g., rash, dermatitis acneiform) common with RAF/MEK inhibitors. **Serious Adverse Events (SAEs):** Standard 24-hour rapid reporting timeline to the sponsor and regulatory authorities. **Data Monitoring Committee (DMC):** Utilized to review safety data during Part 1 of the trial to ensure combinations are safe and have beneficial effects before advancing to Part 2 expansion.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy analyses; Safety Set for all participants who received at least one dose of study treatment. **Sample Size Justification:** Targeted enrollment of 134 participants. The two-part design accommodates patients randomized 1:1:1 to assess safety/preliminary efficacy, appropriately powered to select the optimal combination for Part 2 expansion. **Handling of Missing Data:** Standard oncology imputation; individuals without a post-baseline tumor assessment are typically considered non-responders for ORR.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring visits to ensure protocol adherence and patient safety. **Audit Trail:** Continuous eCRF locking and query resolution via standard sponsor-managed data clarification systems.

11. Study Identifiers & Governance

Primary ID: CLXH254C12201 **NCT Number:** NCT04417621 **Posting ID:** 1035438760563737702 **Sponsor/Lead Organization:** Novartis **Study Title:** Study of Efficacy and Safety of LXH254 Combinations in Patients With Previously Treated Unresectable or Metastatic Melanoma **Official Regulatory Title:** A randomized, open-label, multi-arm, two-part, phase II study to assess the efficacy and safety of multiple LXH254 combinations in patients with previously treated unresectable or metastatic BRAFV600 or NRAS mutant melanoma

12. Clinical Framework (Melanoma Specific)

Disease Area: Melanoma **Primary Condition Name:** Malignant Melanoma (Preferred Name: melanoma) **Preferred UMLS Name:** Advanced Melanoma **Semantic Type:** Neoplastic Process **Clinical Definition:** A malignant neoplasm derived from cells that are capable of forming melanin, which may occur in the skin of any part of the body, in the eye, or, rarely, in the mucous membranes of the genitalia, anus, oral cavity, or other sites. It occurs mostly in adults and may originate de novo or from a pigmented nevus or malignant lentigo. Melanomas frequently metastasize widely, and the regional lymph nodes, liver, lungs, and brain are likely to be involved. The incidence of malignant skin melanomas is rising rapidly in all parts of the world. (*Stedman, 25th ed; from Rook et al., Textbook of Dermatology, 4th ed, p2445*) **Standardized Terminologies:** * **MeSH Code:** D008545 * **HPO Code:** HP:0002861 *

SNOMED ID: 372244006 * **SNOMED Hierarchy:** 1240414004|Malignant neoplasm|Malignant neoplasm (morphologic abnormality); 115223002|Nevus AND/OR melanoma|Nevus AND/OR melanoma (morphologic abnormality); 108369006|Neoplasm|Neoplasm (morphologic abnormality)

13. Study Procedures & Methodology

Management Pathway: Standard oncological RECIST v1.1 imaging and tumor tracking **Education Protocol (HCP):** Investigator training on dose-modification guidelines and AE management (specifically skin and ocular toxicities) **Education Protocol (Patient):** Counseling on oral medication compliance, side effect tracking, and contraceptive requirements **Quality Audit Mechanism:** Centralized data review, imaging quality control, and clinical site auditing

14. Observation & Follow-Up Schedule

Total Duration: Until disease progression or unacceptable toxicity (estimated up to 2-3 years for responders) **Onsite Visit Cadence:** Three times per cycle for Cycles 1 & 2; twice during Cycle 3; once per cycle (Q4W) thereafter **Remote Monitoring Frequency:** Telephone follow-up up to 4 times a year post-treatment **Checklist Review Interval:** Routine medication diary and AE review at every onsite clinical cycle visit

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				